

ChE 382: Unit Operations Laboratory

LIQUID-LIQUID EXTRACTION

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Introduction

Solvent extraction is a widely used separation process in many parts of the chemical industry. It also finds application in the production of pharmaceuticals, environmental engineering, and the minerals industry.

Solvent extraction is often an alternative to distillation. Both are staged processes that depend upon the distribution of key components between two essentially immiscible phases. Both aim to have phase equilibrium between the phases on each of the stages. Hence, it isn't surprising that the analysis of experimental data for distillation and solvent extraction are very similar.

The book by King [1] gives a good discussion on the selection of a separation process for a given application.

This Unit Operation experiment examines the staged extraction of acetic acid between water and mineral oil (white oil) phases. The data obtained allow a first-pass process design for this system.

Theory [1, 2, 3, 4]

Liquid extraction is the separation of the constituent components of a liquid solution by contact with another immiscible liquid. The contacting liquid is called the solvent; and liquid extraction, therefore, is sometimes called solvent extraction. The addition of the solvent creates a new phase partially miscible with the liquid to be extracted. If the constituents of the liquid to be extracted distribute themselves differently between the two liquid phases, a certain amount of separation is achieved and by operating the system as a cascade of stages, the extraction can be enhanced. The liquid to be extracted is called the feed. The solvent-rich product of the operation is called the extract, and the residual liquid from which solute has been removed is the raffinate. The component of feed which distributes itself in both the liquid phase is called the solute.

A single extraction stage consists of a mixer where the feed and the solvent are mixed intimately using an agitator, and a settler where the mixed liquids are allowed to aerate. The lighter liquid will settle on top while the heavier liquid will settle at the bottom and the two phases can be separated. For an ideal extractor, the two liquid phases will be in equilibrium. Settlers have to be designed in such a way that enough residence time is available for the two phases to separate out in two layers.

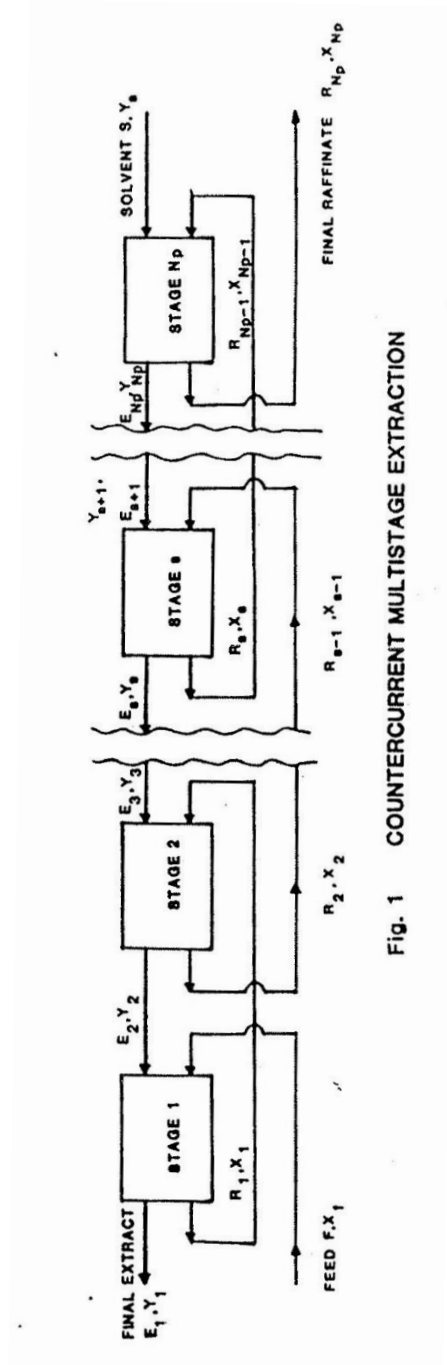


Fig. 1 COUNTERCURRENT MULTISTAGE EXTRACTION

The flow sheet for a countercurrent multistage extraction is shown in *Figure 1*. Extract and raffinate streams flow from stage to stage countercurrently and provide two final products, raffinate E_{Np} and extract E_1 . For a given degree of separation, this type of operation requires less solvent for a fixed number of stages as compared to batch operation. If the flow rate and the composition of feed and solvent are known, the number of stages theoretically required to achieve a desired separation can be calculated graphically, on an appropriate coordinate system (viz. triangular or rectangular) provided equilibrium data for the two immiscible phases are available. The overall efficiency of a cascade with a fixed number of stages can also be calculated.

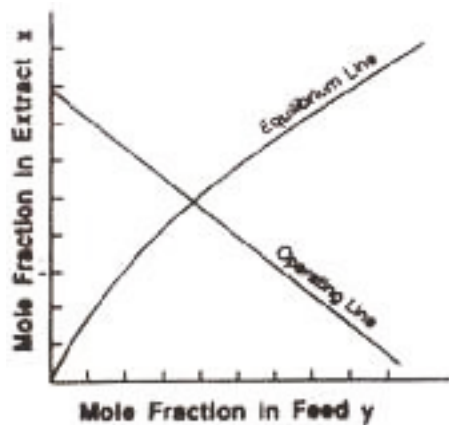
The simplest solvent extraction process is a single stage where the solvent feed containing the component to be extracted is contacted with the extracting solvent. If the phase equilibrium is linear and can be expressed as:

$$x = Ky$$

Where x is the product concentration in the extraction solvent L, y is the product concentration in the feed solvent H, and K is the equilibrium constant.

For the single stage, one can also write a mass balance expressing that the solute extracted into the phase H must equal the solute extracted out of the feed solvent H. One must specify the feed concentration y_F .

The above two equations can be solved to give the equilibrium concentrations x and y . This can also be done graphically as shown below.



The desired solution is the intersection between the operating line (the mass balance) and the line expressing the liquid-liquid phase equilibrium.

This analysis can be extended to a staged system by solving the series of phase equilibria and mass balances for each stage. The solution is simple but tedious algebra. The solution gives the concentration of solute in stage $n+1$ as a function of the initial concentration y_1 .

$$y_{n+1} = \left(\frac{E^{n+1} - 1}{E - 1} \right) y_1$$

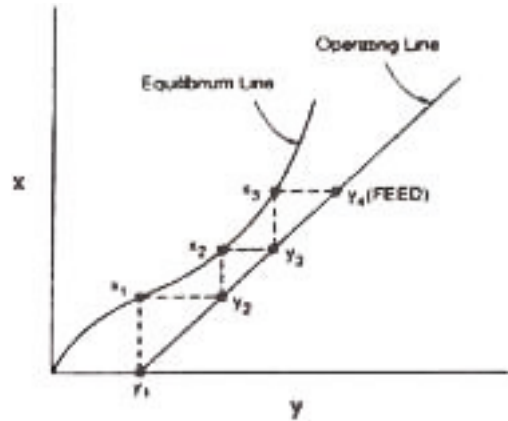
Where E is a constant, defined as:

$$E = \frac{KL}{H}$$

The higher the value of E , the greater the extent of extraction.

This staged extraction can also be solved graphically using an analysis analogous to the McCabe-Thiele treatment of distillation. The plot below shows the equilibrium and

operating curves and the number of theoretical stages is determined by the “staircase” procedure as shown below:



The concept of efficiency can be introduced into solvent extraction in a similar fashion to distillation where the efficiency η is defined as:

$$\eta = \frac{N_A}{N_T}$$

Where N_A is the actual number of stages and N_T is the theoretical number.

With reference to *Figure 1*, one can define various efficiencies as follows:

1. For stage S , Murphree of the extract phase is defined as:

$$E_{ME} = \frac{Y_{S+1} - Y_S}{Y_{S+1}^* - Y_S}$$

Where:

Y_{S+1}^* = composition of extract phase in equilibrium with raffinate phase leaving the $(S + 1)^{th}$ stage.

2. Murphree efficiency of the raffinate phase is defined as:

$$E_{MR} = \frac{X_S - X_{S+1}}{X_S - X_{S+1}^*}$$

Where:

X_{S+1}^* = composition of the raffinate phase in equilibrium with extract phase leaving the $(S + 1)^{th}$ stage.

3. The overall efficiency is defined as:

$$E_O = \frac{\eta_T}{\eta_{act}}$$

Where:

η_T = theoretical number of stages required.

η_{act} = actual number of stages used.

For multicomponent systems and complex processes with recycle and multiple feed, the liquid-liquid extraction can best be analyzed using one of the commercial process simulators. These programs solve the mass balances and phase-equilibrium relations for each of the stages. The solution can be numerically complex due to the highly nonlinear behavior which is often typical of the phase equilibria in liquid-liquid systems. These programs can also take into account the distribution of the two solvents between the phases; something which had to be assumed negligible in the simple graphical analysis.

An understanding of the basic concepts of solvent extraction gained by the simple analysis allows the engineer to intelligently approach complex systems.

Experimental

The experimental unit consists of three stages of solvent extraction, each stage equipped with an agitator to assure good mixing and near equilibrium in the stages. The feed for the extraction should be 0.5 weight percent acetic acid in mineral oil (white oil).

Object

- i. To operate a three stage mixer-settler system to extract acetic acid from white oil using water as the solvent.
- ii. To investigate effect of the speed of mixing on the efficiency of extraction of the unit.

Coordination Between AM and PM Lab Teams

In order to reduce the amount of white oil used during the semester, the white oil from each experiment will be re-used.

Morning (AM) procedure, Pre-lab period: White oil from the previous week should still be in the feed or product tank. If desired, it can be used to set up the oil/water extraction circuit in order to become familiar with the operation of the equipment. At the end of the period, DO NOT DRAIN the oil, but leave it for the afternoon section.

Morning (AM) procedure, Final lab period: The white oil used in the previous experiments will be used for your lab experiments. If the oil is in the product tank, pump it back to the feed tank. Check the level and ask the TA or instructor for help to top off the level to ~25 gallons. You should check whether there is a trace amount of acetic acid in your feed by performing a titration as described below. Then follow the procedure below to make up your acetic acid/oil mixture to the required amount. At the end of your experiments, DO NOT drain the oil, but pump it back to the feed tank for the afternoon lab session. (TA or instructor will help out).

Afternoon (PM) procedure, Pre-lab period: White oil from the morning should still be in the feed or product tank. If desired, it can be used to set up the oil/water extraction circuit in order to become familiar with the operation of the equipment. At the end of the period, DO NOT drain the oil.

Afternoon (PM) procedure, Final Lab Period: The white oil used in the morning experiments will be used for the afternoon lab experiments. If the oil is in the product tank, pump it back to the feed tank. Check the level and ask the TA or instructor for help

to top off the level to ~25 gallons. You should check whether there is a trace amount of acetic acid in your feed by performing a titration as described below. Then follow the procedure below to make up your acetic acid/oil mixture to the required amount. After your experiments are completed, you can pump the oil back to the feed tank; it can be used for the next set of lab teams in the pre-lab to become familiar with the extraction equipment.

Procedure

- i. Study the mixer-settler unit carefully. The Denver system is a miniaturization of a larger scale operating equipment. Each stage contains a mixer and a settler and arrangement of reflux from each stage. The speed of the agitator can be varied by adjusting a knob on the motor. The agitator speed can be measured with a tachometer. There are two rotameters, one each for measuring the flow of the feed and the solvent. The flowrates of extract and raffinate phases can be measured with a measuring cylinder.
- ii. Prepare a mixture of 25 gallons of White Oil and acetic acid so that the concentration of acetic acid is approximately 0.5 weight percent Acetic Acid. Mix them thoroughly in the feed tank by operating the centrifugal pump in closed circuit, and then take a sample for accurate determination of its composition.
- iii. In order to measure composition of acetic acid in either White Oil or water phase, it is titrated against a standardized NaOH solution using phenolphthalein as the indicator. Several concentrations of standardized NaOH solutions are available.
- iv. The solvent circuit (i.e. water) is first established in the mixer-settler unit so that steady levels are reached in each stage. The speed of the agitator is then adjusted to a pre-selected value (see below for agitator speed range).

After establishing the water circuit, start feeding white oil and adjust the flow such that the ratio of solvent to feed flow rates is at least 2.0. Typical flowrates that provide proper operation are: water flowrate of 0.67 GPM and oil flowrate of 0.34 GPM. The separate flows have to be chosen by trial and error to get steady levels of the two phases in each stage. Once the flows have been established, they should be kept constant throughout the experiment. The white oil flowrate must be so chosen that three runs may be conducted with 25 gallons. Allow the system to run until steady state is established. Withdraw samples of white oil and water phases from each of the three stages and determine their composition by titration.

Repeat for a total of three different agitator speeds, keeping the flowrates constant all the time. In all the above steps, keep all the reflux values fully closed.

Cautions

1. Do not start white oil flow unless and until water circuit has been established.
2. Do not run the centrifugal pump in dry condition i.e., the feed tank should always have a minimum level of the liquid.
3. Do not select an excessively high speed ($> 1000 \text{ RPM}$) or too slow speed ($< 300 \text{ RPM}$) for the agitators. An agitator speed greater than 300 rpm is needed to establish the flow circuits of the oil and water in the three stages.
4. Keep the feed tank and all the stages covered to avoid the loss of acetic acid by evaporation except when extracting samples.

Sources of Error

- i. Choice of improper speed of agitation. If the speed is chosen very slow, steady-state levels of liquids are difficult to establish in the stages.
- ii. Measurement of liquid compositions when steady state has not actually been established.
- iii. Wrong preparation of standard solution of NaOH.
- iv. Wrong interpretation of the end point of titration between NaOH solution and the sample.
- v. Improper choice of liquid flow rates.

Data Analysis and Questions

1. Construct plots of acetic acid concentration (or mole fraction) as a function of stage number and agitator speed, for the two phases (oil and water).
2. Plot Y_A (acetic acid in extract) vs. X_A (acetic acid in raffinate) for the different agitator speeds.
3. Perform a McCabe-Thiele analysis to obtain the number of theoretical stages (ideal stages) for the liquid extractor. (See Unit Operations textbook, Ch.23 p.781-786).
4. Calculate Murphree efficiencies for the stages and the overall efficiency.
5. What is the effect of agitator speed on the efficiencies and acetic acid concentration?
6. What changes would you expect if a reflux was used in the extractor?

References

- [1] C. J. King, *Separation Processes*, 2nd ed. McGraw-Hill, 1980.
- [2] W. McCabe, J. Smith, and P. Harriot, *Unit Operations of Chemical Engineering*, 7th ed. McGraw-Hill, 2005.
- [3] R. H. Perry and D. W. Green, Eds., *Perry's Chemical Engineer's Handbook*, 7th ed. McGraw-Hill, 1997.
- [4] P. C. Wankat, *Equilibrium-Staged Separations*. Prentice Hall, 1988.